
ARTICLE

Minimum wages and pass-through

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Abstract

Retail food prices rose dramatically in late 2021. Some argue that this food price inflation was due to “greedflation,” or firms increasing downstream prices simply because they can. In this study, we investigate the sources of “overshifting” store-level cost shocks into downstream prices, or the apparent ability of retailers to pass along price increases that are proportionately larger than increases in cost. We use exogenous changes in minimum wages as our setting and study how food retailers pass increases in labor costs along to consumers in the form of higher food prices. We derive a new theoretical model of retail price pass-through and show that demand curvature, market power, and consumer search behavior each likely affect observed rates of retail price pass-through. Our structural analysis shows that, after controlling for the primary determinants of wage pass-through, market power and demand curvature explain much of the variation in cost pass-through, although general price inflation has an important role in accentuating the rate of minimum wage pass-through. Our findings have important implications for minimum wage policy, and for understanding the role of cost shocks in food price inflation.

KEYWORDS

consumer search, demand curvature, food price inflation, imperfect competition, minimum wages, pass-through

JEL CLASSIFICATION

D43, L13, M31

1 | INTRODUCTION

Minimum wages, at least at the state and local levels, have been rising rapidly for over a decade. For most of that period, food price inflation has been relatively low, averaging some 2.5% over the 2011–2021 period (U.S. Bureau of Labor Statistics, 2023). However, starting in late 2021, food price inflation accelerated rapidly, averaging over 9.0% in the United States from mid-2021 to mid-2023 (U.S. Bureau of Labor Statistics, 2023). Retailing costs rose substantially over the 2021–2022 period, but whether higher food prices are evidence of “greedflation”—defined as increasing prices at a rate greater than the increase in cost, or retailers merely passing higher costs along to consumers—is an important policy question. Changes in labor costs represent an ideal way to examine pass-through, as they are important to retailers’ cost structure (some 17% of operating costs; U.S. Census Bureau, 2023) and represent a change in the cost of selling all products and not just individual categories of products, and some wage changes are exogenous to retailers’ operating environments: minimum wages. Several recent studies examine the pass-through of rising minimum wages to food prices and report mixed findings of either no pass-through at all (Ganapati & Weaver, 2017), relatively small pass-through elasticities (Renkin et al., 2022), or overshifting (pass-through greater than 1; Leung, 2021). Each of these studies, however, examines data from periods in which general price inflation was relatively low and uses reduced-form models that cannot take either imperfect competition or the structure of demand into account in estimating pass-through. In this article, we reexamine the role of minimum wages in retail food prices and explicitly consider the role of generalized price inflation in a structural model of retail price competition and consumer search.

Many firms are only indirectly affected by higher minimum wages, either because they use relatively little minimum wage labor, or wages form only a relatively small part of their overall cost structure. However, pass-through in multi-echelon, differentiated-product supply chains can be substantial, as firms have considerable flexibility in setting prices (Ellickson, 2016; Ellickson & Grieco, 2013).¹ It is well understood that the rate at which firms pass through higher costs to retail prices depends on both the curvature of demand, or how elasticity changes along the demand curve, and the extent of competition among firms in the market (Anderson et al., 2009; Bulow & Pfleiderer, 1983; Hong & Li, 2017; Miller et al., 2017; Miravete et al., 2018, 2022; Weyl & Fabinger, 2013). In the textbook case, with linear demands, perfectly competitive firms pass through any change in cost completely. However, because demand is rarely linear, the degree of competitiveness and the curvature of demand interact to produce ambiguous results (Miravete et al., 2018; Weyl & Fabinger, 2013). With linear demands, constant marginal cost, and constant industry conduct (i.e., the degree of competitiveness does not change with prices), pass-through varies continuously from 100% for firms in perfectly competitive industries to 50% for monopolies (Genakos & Pagliero, 2022; Miller et al., 2017; Weyl & Fabinger, 2013). In a more general empirical setting, this textbook case does not apply, and the effect of market power is indeterminate, depending on the interaction between conduct and curvature.² We account for this interaction in our empirical analysis and consider consumer search behavior.

Anecdotal evidence in the media reflects a common assumption that firms with market power raise prices during inflationary periods simply because they can (Curran, 2023; DePillis, 2022). However, the problem with this sort of casual empiricism lies in the fact that, from the perspective of food markets in 2022, retail concentration has risen relatively little over the past 10 years (McDonald et al., 2023) but food price inflation only emerged in late 2021. What changed to allow food price

¹Further, retail grocery stores are intermediaries, so cost of goods sold (COGS) is clearly their most important cost element (U.S. Census Bureau). However, labor forms most of the remaining variable cost (14%; U.S. Census Bureau) and is subject to greater retailer control than are wholesale prices. Further, some 25% of workers in the grocery industry are within 1% of the minimum wage (U.S. Census Bureau, Current Population Survey), so changes in the minimum wage are likely to apply to a substantial share of grocery workers. Therefore, it is likely that changes in minimum wages have material impacts on retailers’ operating costs (U.S. Census Bureau, 2023, 2024).

²In fact, Miller et al. (2017) show that under log-convex demands (typical with constant elasticity of substitution demands, and possible under almost Ideal demand system), pass-through increases as industry conduct becomes more competitive, which is counter to the “greedflation” story that oligopolies pass through costs at a greater rate than competitive firms.

inflation in 2021? When all prices are rising quickly, consumers face a fundamental signal extraction problem that obscures relative price changes from changes in the overall level of prices (Lucas, 1972). More generally, we argue that consumer search plays an important role in an inflationary environment, as general price inflation increases the dispersion of nominal prices across retailers in the same market (Sara-Zaror, 2024), which leads consumers to search more intensively, demand to become more elastic, retailers to price more competitively, and hence pass-through that is closer to the competitive level (Sara-Zaror, 2024). On the other hand, greater price dispersion may give retailers cover to increase consumer prices without fear of retribution (Ellison & Ellison, 2009; Ellison & Wolitzky, 2012; Richards et al., 2020) in a form of strategic obfuscation.

Greater price dispersion does not imply that consumers do not search. On the contrary, consumers search more with rising prices, but their search is less effective. In fact, it is well understood that pass-through depends on the extent to which consumers search for the lowest price in the market (Byrne & de Roos, 2017; Chandra & Tappata, 2011; Lewis, 2011; Tappata 2009; Yang & Ye, 2008). While most of this literature seeks to explain the “rockets and feathers” phenomenon, or the empirical observation that retail prices tend to rise after a cost shock faster than they fall after the cost shock disappears, their insights have direct relevance for empirical analysis of minimum wage pass-through. Namely, when general prices are rising, Lewis (2011, 423) shows that “for a given level of consumers’ expectations: equilibrium prices increase more quickly with wholesale cost as costs rise relative to expectations.”) Combining the obfuscation and search insights, we argue that inflation causes consumers to search more and have less success doing so, with the result that individual product prices rise more quickly than in low-inflation settings.

Understanding retail food price inflation is more than just a methodological or theoretical curiosity due to the importance of food at home in consumers’ household budgets (Ganapati & Weaver, 2017; Leung, 2021; Renkin et al., 2022). Among the lowest-income households (lowest quintile), consumers spend more than 30% of their income on food at home, largely purchased through grocery stores (Economic Research Service, 2023), whereas consumers closer to the median still spend nearly 10% of their income on food. Although we do not address the aggregate welfare effects of minimum-wage changes, the fact that food purchased at grocery stores represents an important use of income argues for the overall importance of studying this issue, and of getting it right.

We frame our analysis using data on retail food prices. In fact, the retail food industry represents an ideal setting to study the effects of imperfect competition and inflation on minimum wage pass-through, as the degree of competitiveness varies considerably by market (Hosken et al., 2018) and local competition is typically intense (Arcidiacono et al., 2020; Ellickson et al., 2020; Ellickson & Grieco, 2013). Although large national retailers tend to follow relatively uniform pricing strategies, which in absolute terms would mitigate their usefulness in studying pass-through (DellaVigna & Gentzkow, 2019; Hitsch et al., 2021), our data include regional retailers that do not follow national pricing strategies. Whereas others who study minimum wage pass-through in the grocery industry do not account explicitly for competition among retailers (Leung, 2021; Renkin et al., 2022), we examine the role of localized imperfect competition in a structural way.

Others provide extensive empirical evidence on the extent of minimum wage pass-through in the retail industry (Leung, 2021; Renkin et al., 2022), but none address how minimum wage pass-through may change with demand curvature, market power, and generalized price inflation.³ In this article, we test a structural model of minimum wage pass-through and offer a simple explanation for how minimum wage pass-through to retail pricing is likely to vary across different macroeconomic environments, controlling for the degree of competitiveness and curvature of demand.

³There is a deep and unsettled literature on the employment effects of minimum wages (for the seminal article on this topic, see Card & Krueger, 1994; for an early review of the literature, see Neumark & Washcher, 2007), but the weight of the recent empirical evidence suggests that there are very small impacts of an increase in the minimum wage on employment (Allegretto et al., 2017; Azar et al., 2019; Cengiz et al., 2019; Dube et al., 2010; Dustmann et al., 2022), despite orthodox reasoning that suggests the opposite—in the perfectly competitive case, binding minimum wages must have a negative effect and create unemployment at the new minimum wage. Considering this question, however, is related to but beyond the scope of our research.

We use changes in the minimum wage as a clean, exogenous source of econometric identification (Allegretto et al., 2017). Studying wage pass-through to retail prices is very different from studying more usual product-level cost shocks (Lan et al., 2022; Miravete et al., 2018, 2022, 2023). Whereas changes in commodity prices affect the cost of acquiring individual products at the wholesale level, wage changes affect the cost of selling all items more or less equally.⁴ Typical structural models of retail pass-through use matrices of cross-price effects to control for a retailer's internal incentives to change prices in response to changes in acquisition costs, and cross-price matrices among competing retailers to control for the competitive incentives to change prices (Hong & Li, 2017; Richards et al., 2012). In our case, however, changing labor costs only affect a retailer's incentive to change general price levels in competition with other retailers, assuming labor forms a different percentage of operating costs among different retailers. Therefore, we examine how changes in labor cost, induced by changes in the minimum wage, are passed through to consumers at the store level, and not through the structure of individual-product or even individual-category demand.

We find that market power is the most important determinant of minimum wage pass-through, but demand curvature and consumer search are both statistically and economically relevant to the rate of minimum wage pass-through. In general, taking all influences into account, we find that retailers tend to pass higher minimum wages onto consumers more than completely, or overshift costs, at a rate of 140%, meaning that retailers increase prices by 14% for each 10% increase in the minimum wage. With respect to our focal mechanisms, we show that a relatively small change in retail market power (e.g., a 10% increase in an index of conduct that varies from 0 for perfect competition to 1.0 for Bertrand-Nash rivalry) is associated with a pass-through rate that is 70% higher than otherwise. Further, we find that demand curvature, in spite of its theoretical importance in driving pass-through (Miravete et al., 2022, 2023; Weyl & Fabinger, 2013), is a relatively minor contributor to empirical pass-through estimates. Despite the fact that extreme values of curvature have an important influence on pass-through, the fact that curvature varies near 1.0 in our data means that it is of little consequence in reality.⁵ Similarly, a doubling of inflation (e.g., an increase in the rate of inflation from 4.5% to 9.0% witnessed in 2022) is associated with an increase in pass-through of only 3%, or from our observed rate of 140% to 143%. Although statistically significant, the search effect is clearly a relatively minor influence on the observed rate of pass-through.

We make three primary contributions to the empirical industrial organization, labor economics, and policy literatures. First, we contribute to the empirical industrial organization literature by providing a theoretical explanation for an empirical test of the relative importance of market power, demand curvature, and consumer search to the importance of cost pass-through to retail prices (Birchall & Verboven, 2022; Miravete et al., 2018, 2022). Including search in a theoretical model of pass-through reflects the notion that when retail prices rise amid more general price inflation, consumers search more intensively but with less success, so retailers' attempts to raise prices are able to stick and are not competed away. We provide both an empirical test and a theoretical explanation for why this variation on the more general theme of consumer search may be important to the conduct of retail markets (Allender et al., 2021; Ellison & Ellison, 2009; Lewis, 2011). Although others document the importance of accounting for structural attributes of demand while estimating pass-through, we are the first to disaggregate the relative importance of these three structural features on the rate of retail cost pass-through.

Second, we complement recent research in the labor-economics literature by Leung (2021) and Renkin et al. (2022) by providing structural estimates of minimum wage pass-through in the food retailing industry. While both studies discuss the importance of the curvature of demand and competition in affecting the rate of minimum wage pass-through, neither accounts for these features in a way that allows pass-through to vary by the degree of curvature, the intensity of consumer search,

⁴Wages for store workers are variable costs in the sense that selling more goods requires more labor, unlike management wages that enter into a store's fixed costs of operating.

⁵We estimated our empirical model focusing only on high curvature ($\rho > 1$) observations and found substantially higher rates of pass-through. These estimates are available from the authors.

and the extent of competition. Our structural model allows us to directly estimate the impact of all three features on the rate of minimum wage pass-through to retail prices.

Third, we contribute to the policy discussion on the relative importance of “greedflation” as a cause of general price increases throughout the economy. We argue that the notion of greedflation implies that firms increase prices at a rate greater than 100%, or overshift cost increases, simply because they can. We provide empirical evidence that the rate of pass-through depends largely on the exercise of market power in individual retail markets, as the greedflation notion implies, but that structural features such as demand curvature are also important.

In the next section, we describe a simple model of price setting under oligopoly competition, highlighting the importance of demand curvature and consumer search. In Section 3, we explain the data used in our analysis and provide some summary and model-free evidence on the relationship between minimum wages, retail prices, and the general rate of price inflation. In Section 4, we explain the empirical model we bring to the data and show how pass-through rates are likely to be affected by demand curvature, consumer search, and market power exercised by retailers. In Section 5, we present and interpret our results, both in terms of the base rate of pass-through and the relative effect of curvature, market power, and inflation on the estimated rate of pass-through. The final section concludes and offers some policy insights on how retail price pass-through varies over settings that may differ in terms of retail market power, search behavior, and the structure of demand.

2 | IMPORTANCE OF MARKET POWER, CURVATURE, AND SEARCH

Our structural model and counterfactual simulation exercise produce estimates of the minimum wage pass-through rate that vary over time, stores, and markets, which captures variation in demand curvature, the degree of oligopoly market power among retailers, and a way to proxy consumer search behavior. In this section, we explain why each of these elements are important to pass-through by extending the canonical theoretical model of pass-through (Byrne & de Roos, 2017; Genakos & Pagliero, 2022; Miller et al., 2017; Weyl & Fabinger, 2013) to allow for search behavior (Lewis, 2011).

2.1 | Theoretical model of pass-through

Pass-through, or the rate at which increases in cost are passed through to consumers, is critically important and yet frequently misunderstood. Empirical researchers commonly regard market power as the primary determinant of pass-through (for a review of the early literature, see Berck et al., 2009), yet theoretical models of cost pass-through show that the pass-through process is much more nuanced and depends not only on market power, but on the structure of demand (elasticity and curvature) and consumer behavior as well. In this section, we summarize the state of the theoretical art in understanding pass-through and then extend the basic model to account for search behavior.

In general, curvature refers to how the elasticity of demand varies along the demand curve. Intuitively, curvature derives from heterogeneity in consumers' willingness to pay, or the mix of high-willingness-to-pay and low-willingness-to-pay consumers in the market (Miller et al., 2017; Miravete et al., 2018, 2022, 2023). Consider the case of log-convex demands (i.e., demands with “high” curvature). Greater heterogeneity in consumers' willingness to pay implies more curvature and a correspondingly higher rate of pass-through. As costs increase, firms are willing to “give up” low-willingness-to-pay consumers by raising prices in order to exploit the much higher willingness to pay among consumers in the higher end of the market. What appears as something akin to price gouging is simply exploiting the natural curvature of demand (Miller et al., 2017). In this section, we show how this happens more formally and derive testable hypotheses for our empirical model below.

Consider the stylized market of Weyl and Fabinger (2013) that consists of n symmetrically differentiated oligopoly firms. If they each face inverse demand curves $p(q)$, then Weyl and Fabinger (2013) show that joint profit maximization implies an equilibrium condition,

$$\theta = \left(\frac{p(q) - c(q)}{p(q)} \right) \varepsilon_D, \quad (1)$$

where $c(q)$ is the marginal cost of production, $\varepsilon_D = -p/qp'$ is the elasticity of inverse demand, and θ is a conduct parameter defined such that $\theta = 0$ implies competitive behavior, $\theta = 1$ is monopoly pricing, $\theta = 1/n$ is Cournot, and $\theta = 1 + \sum_{i \neq j} \left(\partial q_j / \partial p_i \right) / \left(\partial q_i / \partial p_i \right)$ is Bertrand-Nash conduct in terms of diversion ratios. Further, define the elasticity of supply as $\varepsilon_S = c(q)'(p/q)$; the elasticity of conduct with respect to output as $\varepsilon_\theta = (d\theta/dq)(q/\theta)$, and curvature as the elasticity of marginal surplus with respect to output as $\rho = p_q(q) / (qp_{qq}(q))$. With these elements, Weyl and Fabinger (2013) show the rate of pass-through as a function of the degree of market power, demand curvature, and the structure of cost and demand to be

$$\phi = \frac{1}{1 + \frac{\theta}{\varepsilon_\theta} + \frac{\varepsilon_D - \theta}{\varepsilon_S} + \theta(1 - \rho)}, \quad (2)$$

where ρ is the measure of demand curvature in Miravete et al. (2022). In this specification, it is straightforward to recognize that under assumptions of constant marginal cost ($\varepsilon_S = 0$) and constant monopoly conduct ($\theta = 1, \varepsilon_\theta = 0$), linear demands ($\rho = 0$) imply a pass-through rate of 1/2 and competitive conduct implies complete pass-through ($\phi = 1$) as the “textbook” benchmarks for pass-through. Extending this intuition further, if $\rho > 1$, then demand is log-convex, the denominator of (2) is smaller, all else constant at the monopoly values, and firms pass through cost increases at a rate greater than 100%, or costs are overshifted. In the concave-demands case ($\rho < 0$), the denominator in (2) is larger, pass-through is less than 100%, and firms absorb some of the cost increase as lower margins. Essentially, curvature interacts with market power to accentuate its effect on pass-through in the log-convex demands case, raising the rate of cost pass-through and creating the possibility that retailers overshift cost increases into consumer prices, and profit as a result.

We extend the basic model of Weyl and Fabinger (2013) to include consumer search and the role of inflation in pass-through. When consumers actively search, or choose whether to search or not search, rational retailers play mixed strategies in prices (Byrne & de Roos, 2017; Chandra & Tappata, 2011; Lewis, 2011; Tappata, 2009; Yang & Ye, 2008). In a food-retailing context, mixed strategies in individual product prices manifest as “temporary price reductions” or sales (Varian, 1980) in which the retailer maintains relatively high everyday prices, but attracts searching consumers by periodically reducing prices below their usual level. When the market consists of a distribution of searching and nonsearching consumers, Lewis (2011) shows through numerical simulations of his mixed-strategy equilibrium that retail prices rise more quickly in wholesale prices (i.e., pass-through rises) when consumers’ expectations of wholesale prices are below their current actual level. This is precisely the situation when inflation rises unexpectedly. Consumers search more when prices are rising, so pass-through rises in inflationary periods.

Incorporating search directly into a structural expression for equilibrium pass-through is complicated by the fact that the Weyl and Fabinger (2013) result considers *pure* strategies, whereas consumer-search arguments typically assume firms compete in *mixed* strategies. Therefore, we summarize the implications of consumer search by incorporating a price dispersion parameter, $T(q)$, in

the equilibrium condition of Weyl and Fabinger (2013).⁶ From Tappata (2009) and Lewis (2011), we know that $S(p)$, the distribution of searchers across prices, is strictly increasing in p . As prices rise, the payoff to search increases, incentivizing more consumers to search. Because $S(p)$ is monotonic in prices, we invert the distribution of searchers so that $S^{-1}(p) =: T(q)$, which we define to be the number of searchers in terms of *quantity* instead of prices. When we add $T(q)$ in the equilibrium condition of Weyl and Fabinger (2013) and derive an expression for pass-through, we find

$$\phi = \frac{1}{1 + \frac{\theta}{\varepsilon_\theta} + \frac{\varepsilon_D - \theta}{\varepsilon_S} + \theta(1 - \rho) - \frac{\varepsilon_D}{\varepsilon_T} \frac{T'(q)}{p(q)}}, \quad (3)$$

where $\varepsilon_T = \frac{T'(q)}{T(q) \cdot q}$ is the search elasticity.⁷ The only difference between the outcome for pass-through in (2) and (3) is the negative term $-\frac{\varepsilon_D}{\varepsilon_T} \frac{T'(q)}{p(q)}$ that results from incorporating search. Expression (3) implies that as search intensity (in this case, search elasticity ε_T) increases, pass-through (ϕ) also increases because ε_D is negative by definition. As we explain more completely below, we do not observe search behavior in retail markets, so we proxy consumer search behavior with its cause: the rate of general price inflation. Consequently, as inflation rises, the number of searchers rises, and we expect pass-through to increase accordingly.

While the market power and search elements in Equation (3) follow from our equilibrium pricing model below, it remains to demonstrate more carefully exactly how empirical models of demand are able to identify variation in curvature that are clearly important to empirical estimates of pass-through. We do so in the next subsection.

2.2 | Accounting for curvature

From the theoretical model of pass-through developed in the last subsection, the curvature of demand is clearly of primary importance. In our empirical application, we follow others in the retail-store demand literature (Bell & Lattin, 1998; Briesch et al., 2009; Briesch et al., 2013) and estimate a discrete choice model of store demand, but curvature is rarely the focus when estimating demand models of this type. Miravete et al. (2022, 2023), however, show that demand curvature is easily and most effectively recovered through the interaction between consumers' price sensitivity and the distribution of unobserved heterogeneity in demands. Their primary result holds that even though most discrete choice models of differentiated product demand allow for the interaction between distributions of attribute preference and consumer heterogeneity (Berry et al. [BLP], 1995; Nevo, 2001), it is really only the interaction between price response and income that matters when curvature is important. In our model, we extend their approach to include search behavior as a means of testing its influence on demand, and ultimately equilibrium prices. Because their approach is somewhat non-standard, we repeat their main finding here as a means of showing how we include search behavior in an empirically tractable model of demand and pricing.

Suppose the utility of consumer i choosing item j in market t is given by (dropping market subscripts for clarity)

$$u_{ij} = \mathbf{X}_j \beta_i + f_{ij}(p_j, \pi, y_i; \alpha_i) + \xi_j + \varepsilon_{ij}, \quad (4)$$

⁶In Weyl and Fabinger (2013), the equilibrium condition that generates the pass-through expression (2) is $p(q) - \theta \cdot ms(q) = mc(q) + a$, where the left-hand side indicates the marginal revenue given conduct parameter θ , and the right-hand side indicates the marginal cost plus an exogenous shock (e.g., per-unit tax or minimum wage hike in our case) that shifts the marginal cost curve upward. In the online appendix, our derivation assumes the cost shock derives from an increase in minimum wages (mw) and not a tax.

⁷See the online appendix for full derivation of the pass-through formula involving search intensity.

where $\mathbf{X}_j$ is a vector of attributes with parameters β_i , y_i is the income for consumer i , π is the rate of general price inflation that varies over markets, ξ_j are factors that affect the demand for item j but are unobserved to the econometrician, and ε_{ij} are independent, identically distributed Type I extreme value errors.⁸ Further, allow f_{ij} to be a general sub-utility function that allows for a general set of relationships between price, inflation, and income with the marginal utility of income parameter defined as α_i . As is well understood, the BLP model captures the importance of income in a way that captures the fact that income is likely to be important for higher-priced items, such as among the cars in their empirical example. With our Type I extreme value distributional assumption, the logit choice probabilities are

$$P_{ij}(\mathbf{p}) = \frac{\exp\left(\mathbf{X}_j\beta_i + f_{ij}(p_j, \pi, y_i; \alpha_i) + \xi_j\right)}{\sum_{k=0}^J \exp\left(\mathbf{X}_k\beta_i + f_{ik}(p_k, \pi, y_i; \alpha_i) + \xi_k\right)}. \quad (5)$$

Market demand for item j (in our case, store) is found by integrating over the distribution of consumer heterogeneity, $G(i)$, such that $Q_j(\mathbf{p}) = \int_{i \in I} P_{ij}(\mathbf{p}) dG(i)$. Define $f'_{ij} = \partial f_{ij}(p_j, \pi, y_i) / \partial p_j$ and $f''_{ij} = \partial^2 f_{ij}(p_j, \pi, y_i) / \partial p_j^2$ so that the elasticity of demand is

$$\varepsilon_{D,j}(\mathbf{p}) = \frac{-p_j}{Q_j(\mathbf{p})} \int_{i \in I} f'_{ij} P_{ij}(\mathbf{p}) (1 - P_{ij}(\mathbf{p})) dG(i), \quad (6)$$

where $f'_{ij} = \alpha_i$ in the linear sub-utility case, but more general for other nonlinear sub-utility specifications. Using this expression to show how elasticity changes in the quantity demanded, we arrive at an expression for the curvature (ρ_j) implied by a discrete choice demand system. In this way, curvature includes both the dispersion of demand over the distribution of heterogeneity in (6) and its skewness. Differentiating Equation (6) yields

$$\rho_j(\mathbf{p}) = \frac{p_j^2 Q_j(\mathbf{p})}{\varepsilon_{D,j}(\mathbf{p})} \left[\int_{i \in I} f''_{ij} P_{ij}(\mathbf{p}) (1 - P_{ij}(\mathbf{p})) dG(i) + \int_{i \in I} (f'_{ij})^2 P_{ij}(\mathbf{p}) (1 - P_{ij}(\mathbf{p})) (1 - 2P_{ij}(\mathbf{p})) dG(i) \right] \quad (7)$$

where the term $P_{ij}(\mathbf{p}) (1 - P_{ij}(\mathbf{p})) (1 - 2P_{ij}(\mathbf{p}))$ in the second part of (7) is the skewness of $G(i)$. In the linear sub-utility case, the first term on the right side of (7) is equal to zero, but not the second term, so even in the simplest case, the mixed logit model has the potential to capture a relatively wide range of curvature values.

In fact, Miravete et al. (2022) show how simple logit models restrict the value of $\rho < 1$, restricting pass-through to be less than complete by construction. Further, they demonstrate through Monte Carlo experiments that standard mixed logit models of the form estimated in BLP and Nevo (2001) relax this restriction only marginally. Rather, in order to remove the curvature restrictions on discrete choice models of demand, the distribution of price sensitivity should be a function of income and thereby reflect the underlying asymmetric distribution of income in the population.

In our empirical exercise below, we follow their example and estimate a discrete choice model of retail store demand in which we allow for the price sensitivity of store demand to be a flexible distribution of household income and the general rate of inflation to capture search behavior. In our structural model of retailer pricing, where equilibrium prices reflect the full matrix of own- and

⁸Note that we adopt the generic term “market” from BLP and Nevo to refer to combinations of geography and time period, so each county-week in our data is a different value of t .

cross-price elasticities between competing stores, this demand flexibility is manifest in the fact that elasticities vary over time and among markets in a way that implicitly captures a wide range of curvature values. Because minimum wages also vary over time and across markets, equilibrium prices will therefore reflect not only demand curvature, but variation in competitive behavior (market power) and consumer search (inflation) as well. Therefore, our empirical application tests three hypotheses that follow from the theoretical framework developed above: (1) pass-through of minimum wages to retail prices will rise in the curvature of demand, (2) pass-through will rise in the degree of market power exercised among retail rivals, and (3) consumer search, as proxied by the rate of general price inflation, will also be positively related to the rate of minimum wage pass-through. In the next section, we describe how we measure each variable involved in our estimation procedure more carefully.

3 | DATA AND IDENTIFICATION

3.1 | Data and data sources

We use NielsenIQ retail scanner data from the James M. Kilts Center for Marketing at the University of Chicago Booth School of Business (hereafter, NielsenIQ data) for our primary pass-through analysis, for a time period that includes both low inflation (2011–early 2020) and high inflation (late 2020–2021). In general, choosing a sample time period that includes 2021 exploits a rather unique opportunity to examine retail pricing behavior during a period of high general price inflation—an opportunity that has not arisen in several decades (Figure 1).

The NielsenIQ data are widely used for research in applied demand analysis, in the empirical industrial organization literature, for studying wage pass-through as we do here (Leung, 2021). The NielsenIQ data measure prices, volume, and store merchandising for every item in retail stores (at the universal product code [UPC] level) in every market in the United States, covering some

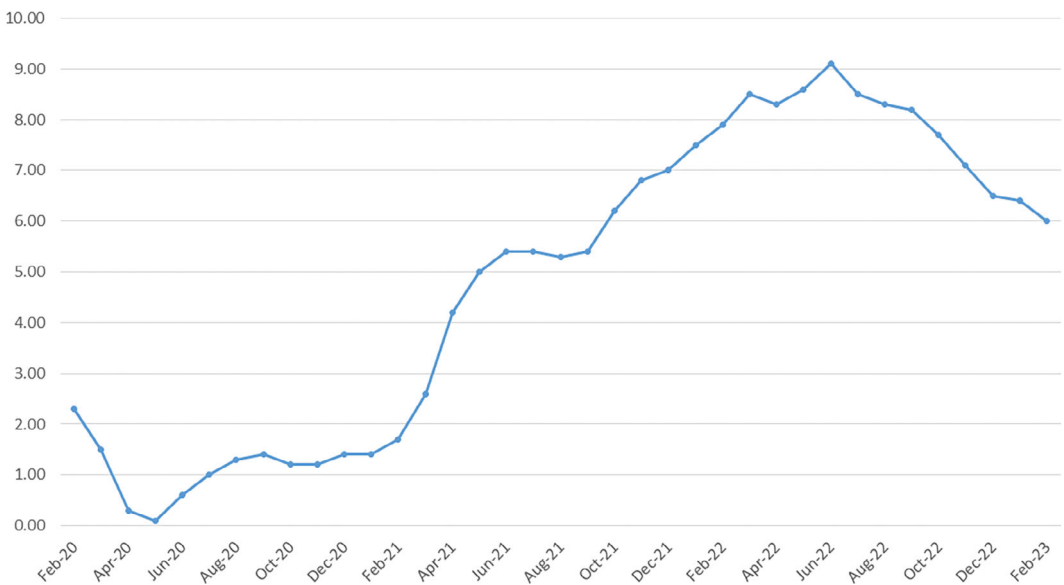


FIGURE 1 Rate of inflation in the United States by month, 2020–2023. Data are from the St. Louis Federal Reserve Board and represent the annual percentage change in the U.S. Bureau of Labor Statistics Consumer Price Index each month of the period represented in the figure.

30,000 – 35,000 total stores. We conduct our analysis at the level of the retail chain, or banner, in each of our sample markets to capture oligopoly rivalry between chains within the same local market. We develop store-level price and quantity indices across all grocery categories for each retail grocery store in the NielsenIQ data, which we explain in more detail below, and aggregate up to the banner level to produce a market-week data set that varies across states and local areas, as well as over time. Our empirical approach, therefore, assumes that consumers choose grocery stores on the basis of basket competitiveness, assuming that all product categories in the store contribute to the attractiveness of each store for consumers' store choice decisions. We treat each county as a market for analytical purposes and consider all the interaction in demand among all stores in each county, grouping stores by ownership group as competing with all other stores in all other ownership groups in each county.

The choice of county as our geographical unit of analysis is not without consequence, or alternatives. Among recent research that uses the NielsenIQ data, defines markets at the designated market area (DMA) level to better understand retailer market power in input markets, while Döpper et al. (2022), Triggs (2021), and Luo (2024) also use DMA-level data aggregations for different purposes.⁹ On the other hand, Kroft et al. (2020) define product markets at the county level to use the state-border-county identification strategy developed by Dube et al. (2010). Because our treatment occurs at the state and/or local levels, and we require accurate measures of market-level income, we follow Kroft et al. (2020) and define markets at the county level.¹⁰

Our data measure store sales and prices on a weekly basis from the first week of January 2011 through the final week of December 2021. We do not aggregate out to the monthly level as others in this literature do (Leung, 2021; Renkin et al., 2022) in order to focus our attention on the announcement effects associated with minimum-wage legislation, and to match the frequency with which retailers change prices in response to their competitive environments and consumer demand (DellaVigna & Gentzkow, 2019; Hitsch et al., 2021; Nakamura et al., 2011). Due to the exceptional nature of retail sales during the COVID-19 pandemic, we control for the onset and development of the COVID-19 pandemic using a full set of weekly controls in our empirical model. Our identification strategy, therefore, compares weekly pass-through over a long pre-inflationary period (before July 2021) with pass-through over several weeks during strong retail price inflation. Our empirical analysis with these data includes both reduced-form models (similar to Leung, 2021, and Renkin et al., 2022) and structural models intended to test our search, curvature, and market power hypotheses.

Beginning with a sampling frame of the entire United States, we remove several markets from our analysis for various reasons. First, because we know that competitive interactions among retailers are likely to be important in determining pass-through rates (Miller et al., 2017; Miravete et al., 2018, 2022; Weyl & Fabinger, 2013), we use data from specific markets in each state and select competing retailers from those markets. Treatments in our data are at the state and/or local area level, so we choose the largest county, by population, in each state as representative of the largest market for groceries. While this approach ignores some important markets in each state, our intent is to sample markets with competing retailers and not necessarily to include them all. Focusing on competitive interactions, however, means that we drop some markets from our analysis that do not report at least two competing retailers that interact over the entire sample period. We realize that all the largest counties in the United States have at least two retailers, but not all report their data to NielsenIQ. Our set of competing retailers, therefore, is conditioned on the subset that report scanner data to NielsenIQ.¹¹ Our final sample, therefore, consists of markets in seven states: Arizona,

⁹DMA's are also known as "media markets," so NielsenIQ reports scanner data by DMA in order to reflect television and radio markets that may be relevant to retailers and product suppliers.

¹⁰There are also 12 DMA's in the United States that split counties, so allocating demographic variables collected at the county level is more arbitrary than we are willing to accept.

¹¹In selecting markets this way, we do not believe we invite sample selection bias, as we chose markets on the basis of data availability only, and not with a priori notions of which are likely to be more or less competitive. By following the procedure described here, we mechanically choose all markets that have at least two rival stores in the data over the sample period. Including single stores, on the other hand, would induce bias, as our pricing model would infer they are monopoly markets when, in fact, they are not.

California, South Dakota, North Carolina, Washington, West Virginia, and Vermont. Second, we retain one state that does not have a minimum wage above the federal minimum (North Carolina) to serve as a control market.¹² Even though there were other markets that could serve as a control market, in the sense that their minimum wage did not vary over the sample period, none also met the criteria of having at least two retailers competing consistently over the entire sample period.

We form our price indices following Beraja et al. (2019) and Leung (2021), with a similar goal of approximating the Bureau of Labor Statistics' Consumer Price Index (CPI) as closely as possible. We assume retailers make pricing decisions at the national or regional level, so we aggregate sales over stores under the same retail banner in the same market to define our price indices.¹³ In fact, we design our sample around the implicit assumption that retailers price consistently across stores in a particular region, based on empirical evidence that there is far more variation in retail prices between chains than within over a national or regional sample (DellaVigna & Gentzkow, 2019; Hitsch et al., 2021).

We assemble our price data as follows: First, we remove any store that does not appear in the data over the entire sample period in order to remove the effect of store entry and exit. Second, others aggregate the data to the monthly or quarterly level to minimize the effect of "chain drift," but at the expense of aggregation bias due to averaging over retailers' pricing cycles.¹⁴ Because retailers compete on a regional basis over weekly pricing cycles, we instead use weekly prices and allow item weights to reflect week-to-week changes in item sales. Third, we construct product group average prices by dividing the total sales in each product group, for each store, by the total number of units sold in the product group. As Leung (2021) explains, the result is a quantity-weighted weekly average price at the product group level. Fourth, we then aggregate product group-level price indices up to the store level using store-level product group shares as weights to construct weekly store price indices. Finally, we aggregate to the chain level by constructing weighted averages over each retail chain, where weights are defined as total store sales divided by chain sales, under the same retail chain in the same market.

In the structural model, which we describe in more detail below, we estimate the choice probability for each retailer, in each market, each week. To estimate the relative attractiveness of each retailer, we measure retailer-week-county market shares from observed dollar-sales of each store, aggregated to the retailer level in the NielsenIQ data. Retailer attraction also depends on marketing mix elements that vary by retailer, and over time (Bell & Lattin, 1998; Briesch et al., 2009; Briesch et al., 2013). Therefore, we also calculate the size of the assortment offered by each retailer by taking a weighted average of the assortments offered in each store, measured as the total number of UPCs sold in each week, with weights defined as store sales divided by total weekly sales of the retail chain. We introduce two marketing-mix elements by calculating the weighted-average probability that a product is on display in each store, and the average feature probability, with weights defined as above. Of course, store choice also depends on the store-level price index.

Demand curvature depends critically on observed heterogeneity, specifically household income in each market (Miravete et al., 2018, 2022, 2023). Our demographic and socioeconomic data are from the NielsenIQ HMS (Household Panel) data set for matching markets and retailers as our RMS scanner data. We use contemporaneous estimates from 2011 to 2021 for average age, education (years attained), average income, and household size for each county represented in our scanner data. Although there are many different potential measures of income, we use household income as

¹²Although our approach does not use a reduced-form difference-in-difference (DiD) structure to identify pass-through rates, we follow others in this literature by recognizing the importance of fundamental identification in structural econometric models (Villas-Boas et al., 2021) by including a control market. The federal minimum wage is currently \$7.25 per hour, so retaining only North Carolina means that we drop 20 states from the analysis. Five states (Alabama, Louisiana, Mississippi, South Carolina, and Tennessee) do not have state minimum wages, Georgia and Wyoming have minimum wages that are below the federal minimum, and the other 13 states have minimums equal to the federal minimum (U.S. Department of Labor, 2023).

¹³Retail banners are not synonymous with ownership groups, as some owners may have several different retail brands, each of which is managed somewhat independently of the others.

¹⁴Chain drift is defined as the difference between a "chained" price index, or one that updates the item weights of each item in the hypothetical consumer basket due to price-induced substitution, and one that uses fixed item weights, such as the BLS CPI. Fixed-weight indices overstate changes in the actual cost of living as consumers substitute among items as relative prices change.

the measure that is most relevant for household-level grocery purchases, aggregated up to the county level by taking a simple average over all households in the relevant HMS data. Table 1 summarizes all our estimation data, including the market-level demographic data and the week-market-banner price indices that form the basis of both our reduced-form and structural models.

We obtain minimum wage data from Vaghul and Zipperer (2016) and define the effective date as the date of implementation rather than enactment. Table 2 shows how minimum wages varied

TABLE 1 Summary of variables used in estimation.

Variable	Units	Mean	Std. dev.	N
Store volume	Units/week	5,627,097	6,717,335	17,343
Feature	%	6.4802	8.5233	17,343
Display	%	4.4530	10.3887	17,343
Variety	#	11,402	7982	17,343
Store price	\$/unit	\$3.5575	\$1.0113	17,343
Number of stores	#	129.0616	249.7131	17,343
Education	Years	11.6379	0.4350	17,343
Income	\$000/years	\$66.5033	\$7.5478	17,343
Age	Years	50.8081	1.4483	17,343
Household size	#	2.2116	0.1183	17,343
Minimum wage	\$/hour	\$9.2339	\$1.8642	17,343
Retail wage	Index	382.5384	23.8377	17,343
Electricity	Index	145.8650	19.2558	17,343
Marketing cost	Index	136.3880	3.9734	17,343
Transportation	Index	139.5035	11.2508	17,343
Food mfg. wage	Index	202.3916	9.1223	17,343
U.S. inflation	%	2.0626	1.3609	17,343

Note: Retail sales and pricing data are from NielsenIQ data on an individual store level. Demographic data are from NielsenIQ HMS data for matching stores and markets. Instrumental variables and cost indices are from the U.S. Bureau of Labor Statistics.

TABLE 2 Minimum wages by year and sample market.

Year	Phoenix, AZ	Los Angeles, CA	Charlotte, NC	Sioux Falls, SD	Burlington, VT	Seattle, WA	Charleston, WV
2011	7.35	8.00	7.25	7.25	8.15	8.67	7.25
2012	7.65	8.00	7.25	7.25	8.46	9.04	7.25
2013	7.80	8.00	7.25	7.25	8.60	9.19	7.25
2014	7.90	9.00	7.25	7.25	8.73	9.32	7.25
2015	8.05	9.00	7.25	8.50	9.15	9.47	8.00
2016	8.05	10.00	7.25	8.55	9.60	9.47	8.75
2017	10.00	10.50	7.25	8.65	10.00	11.00	8.75
2018	10.50	11.00	7.25	8.85	10.50	11.50	8.75
2019	11.00	12.00	7.25	9.10	10.78	12.00	8.75
2020	12.00	13.00	7.25	9.30	10.96	13.50	8.75
2021	12.15	14.00	7.25	9.45	11.75	13.96	8.75

Note: Minimum wage data are from Vaghul and Zipperer (2016), and we extend their data set through 2021 for all markets/states. All wages are in \$/hour and do not include tipped minimums. Charlotte, North Carolina, is included as a control market, as its minimum wage is the federal minimum.

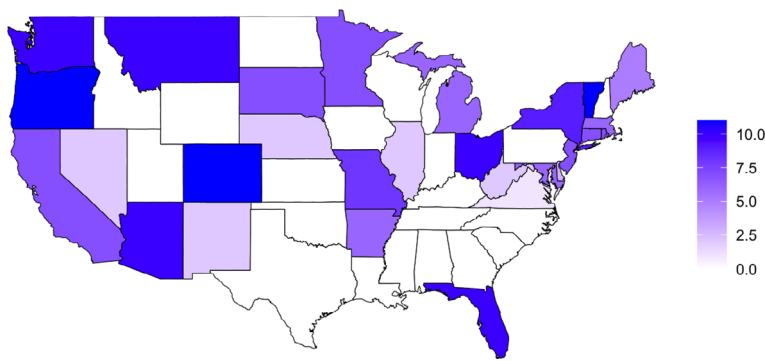


FIGURE 2 Number of state-level minimum wage changes, 2011–2021. The map shows U.S. states with the number of new minimum wage implementations from 2011 to 2021. The graph shows substantial variation in frequency of minimum wage changes in several states, whereas about half of the states do not have minimum wage changes in our sample period. Minimum wage data are from Vaghul and Zipperer (2016).

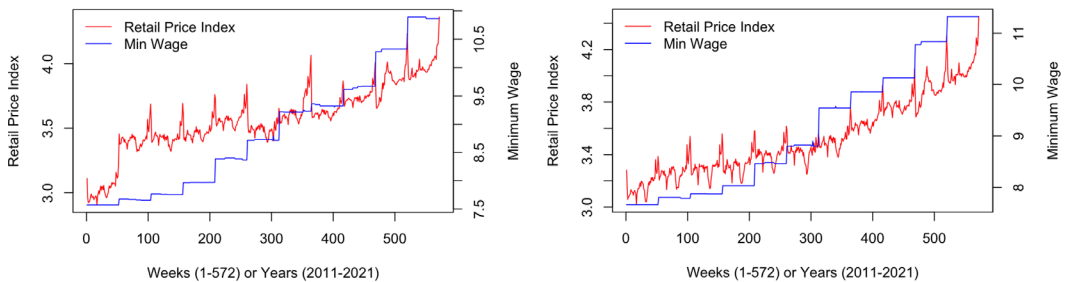


FIGURE 3 Evolution of retail price index and minimum wages, 2011–2021. The first panel shows the evolution of averages of retail price index and minimum wages across our sample period (2011–2021 or weeks 1–572) in all 25 states, which is our entire data set in reduced-form analysis. The second panel shows the evolution within the seven-state data set, which is the data used in structural analysis consisting of a consistent set of at least two competing retailers in each state or federal information processing standard. Both panels show a positive relationship between minimum wages and retail price index. Minimum wage data are from Vaghul and Zipperer (2016), and we compute retail price indices from NielsenIQ/Kilts data.

across our sample markets during the 2011–2021 time period, and Figure 2 shows how minimum wages vary more generally across all markets in the United States. Unlike earlier sample periods used in Leung (2021) and Renkin et al. (2022), minimum wages varied substantially both over time and across the cross-sectional markets in our data.¹⁵ Figure 3 provides summary evidence of the extent of covariation between minimum prices and wages across every state in the United States.

Our inflation data are from the Federal Reserve Board of St. Louis (FRED, 2023) and represent monthly changes in the consumer price index (CPI) calculated by the U.S. Bureau of Labor Statistics. Although this definition of inflation invites criticisms that inflation is endogenous to minimum wage pass-through, retail food represents only 14.2% of the CPI (Statista, 2024), so changes in retail food prices have a relatively small effect on our measure of inflation. Regardless, we conduct a robustness

¹⁵Many of the states in our sample index changes in minimum wages as a matter of policy, for instance, to rise on January 1 of each year in fixed increments. This fact does not materially affect our identification strategy, as retailers price on weekly cycles and still choose whether to pass the indexed minimum wage changes along to consumers based on their competitive situations, as well as their perceptions of consumer demand.

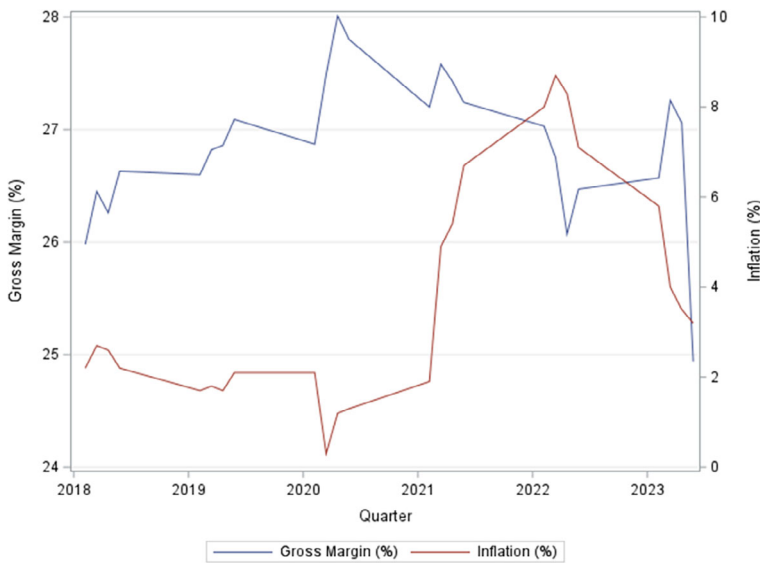


FIGURE 4 Inflation and gross margin, U.S. grocery retailers, 2018–2023. Inflation data are from the St. Louis Federal Reserve Board, and gross margin data are calculated as the average over the top 10 publicly traded grocery retailers in the United States, from Yahoo Finance, on a quarterly basis.

check of our main results with a measure of the CPI that excludes food, and our findings remain qualitatively the same.

This is an important point because our objective is to study not only the pass-through of minimum wages to retail prices, but also how the level of general price inflation affects pass-through. The popular press coined the term “greedflation” as inflation rose during 2022 and firm profits continued to rise, citing higher consumer demand and a seeming unlimited willingness to pay after emerging from the COVID-19 pandemic (DePillis, 2022). In Figure 2, we examine this question visually by showing a summary relationship between quarterly gross margins of all publicly traded food retailers, which includes the retailers in our data, again relative to the general level of price inflation. This graphic shows that there seems to be no credence to this argument, at least in summary data. In fact, there seems to be a negative correlation between inflation and gross margins. Rising profits when all costs are also increasing, however, implies a very general sort of overshifting that has yet to be documented in the empirical literature. In our empirical analysis below, we disentangle consumer and competitive responses and test whether the summary insights in Figure 4 hold up under deeper statistical analysis.

3.2 | Identification and reduced-form estimates

Our identification strategy for the underlying minimum wage-price index relationship is similar to that used in Leung (2021) and Renkin et al. (2022); namely, conditional on a set of controls and fixed effects at the week (they use quarterly and monthly effects, respectively) and store level, stores in areas that did not experience an increase in the minimum wage serve as controls for stores in areas that did.¹⁶ Our strategy is widely accepted in the literature that studies the impact of minimum wages on employment and, as such, is not controversial (Allegretto et al., 2017; Azar et al., 2019;

¹⁶We refer generically to “areas” to include both states and localities. There were no changes in the federal minimum wage over our sample period, but many of the minimum wage changes took place at the substate or locality level.

Cengiz et al., 2019; Dustmann et al., 2022). Given that our main approach is structural, however, we do not rely solely on the causal nature of our findings for our main contribution, but as a means to ensure that our structural findings can indeed be interpreted as causal.

In this subsection, therefore, we offer an econometric analysis of minimum wage pass-through roughly equivalent to Leung (2021) and Renkin et al. (2022) in the recent empirical literature in order to establish a baseline for our structural estimates below, and to compare our estimates to others. For this purpose, we estimate a series of reduced-form models in which we include retailer and week fixed effects, as well as retailer and retailer-county trends, to account for factors that may affect pass-through that are either retailer- or week-specific and yet not otherwise captured by our controls. We also account for the effect of inflation in a straightforward way—simply by interacting the current rate of inflation with changes in the minimum wage.

Our estimates in Table 3 find a direct pass-through rate of 1.8% in the most saturated model (Model 5, with retailer-county trends), which is slightly smaller than Renkin et al. (2022, 3.6%) and also smaller than Leung (2021, 7.0%), and considerably smaller than other estimates in Model 2 (with retailer-only trends).¹⁷ More importantly, however, our estimates in Table 3 show that the pass-through rate in Model 3 is substantially higher in inflationary regimes than otherwise. Because these estimates are in log-values of the price and minimum wage variables, they imply that the base pass-through rate of 1.8% in Model 3 is over 3.4% when inflation interaction is taken into account, which is nearly in line with Renkin et al. (2022) in monthly data. Leung (2021) shows that a pass-through rate of about 7.0% is consistent with complete pass-through, given the share of labor in retailers' cost structures, so our total estimate of 3.4% suggests that retailers still significantly undershift minimum wage increases during inflationary regimes. However, our undershifting estimate is not statistically significant when we allow for retailer-county trends in Model 5, so there is some question as to whether this outcome is likely to hold in a more complete model that takes into account the equilibrium nature of retail pricing outcomes. Moreover, because this is a reduced-form estimate, we cannot examine whether over-overshifting is due to “greed” or the structure of demand and competitive interactions.

That is, these estimates do not test any underlying pass-through mechanism directly. The structural approach of Miravete et al. (2022), on the other hand, allows us to test directly whether pass-through depends on the elasticity and curvature of demand, consumer search behavior, or market power. We turn to this question in the next subsection after explaining how we identify demand manifolds in our retailer data.

3.3 | Structural identification

As is common in structural models, the primary barrier to identification is the endogeneity of prices (BLP; Berry & Haile, 2016). For our structural analysis, we follow Gandhi and Houde (2019) and Miravete et al. (2022) and include distances between fitted prices as instruments in an IV-estimation framework. Namely, we recognize that the goal in instrumenting for prices is to remove endogeneity that derives not only from demand, but from the exercise of market power as well. Gandhi and Houde (2019) argue that if we control as nearly as possible for all of the elements that define the competitive environment of each retailer (in our case) and remove those elements from the inverse demand for the product, then we can isolate the remaining purely exogenous variation in prices. Formally, we write the inverse demand for retailer demand as

$$\phi_{jt}^{-1}(s_t, x_t, p_t; \Psi) = h(\omega_{jt}; \Psi) + C_t(\Psi), \quad (8)$$

¹⁷For interpretation, this 1.8% pass-through rate applies to a 100% increase in minimum wages, or a pass-through elasticity of 0.018. We use the former interpretation throughout for clarity.

T A B L E 3 Reduced-form estimates of minimum wage pass-through: Complete data set.

	Model 1		Model 2		Model 3		Model 4		Model 5	
	Est.	Std. err.	Est.	Std. err.	Est.	Std. err.	Est.	Std. err.	Est.	Std. err.
ln(Min wage)	0.0609	0.0897	0.0686	0.1136	0.0104	0.1166	0.0264	0.1245	0.0180	0.0997
Inflation × Min wage	0.0449	0.0338	0.0359	0.0404	0.0312	0.0363	0.0327	0.0345	0.0163	0.0371
ln(Age)	−0.1085	0.1003	−0.1154	0.0728	−0.2036	0.1331	−0.2086	0.1446	−0.1414	0.0974
ln(Income)	0.0743	0.0595	0.0485	0.0516	0.1219	0.1005	0.0769	0.0950	0.0979	0.0655
ln(Education)	−0.1115	0.2084	−0.1233	0.1326	−0.2133	0.1905	−0.1657	0.1667	−0.0885	0.0878
ln(HH size)	0.0292	0.0587	0.0762	0.0557	0.0374	0.0512	0.0723	0.0421	0.1105***	0.0502
Num. competitors	−0.0010	0.0036	−0.0023	0.0037	−0.0024	0.0035	0.0005	0.0055	−0.0031	0.0043
Constant	1.7478***	0.7635	1.8117***	0.5940	2.1982***	0.7747	2.1490***	0.6414	1.4883***	0.3273
R ²	0.9728		0.9838		0.9706		0.9731		0.9806	
RSS	115.48		68.71		124.85		114.1586		82.25	
Controls	Yes		Yes		Yes		Yes		Yes	
Week FE	Yes		Yes		Yes		Yes		Yes	
Retailer FE	Yes		Yes		No		No		No	
Retailer trends	No		Yes		No		No		No	
Ret.-county FE	No		No		Yes		Yes		Yes	
County trends	No		No		No		Yes		No	
Ret.-county trends	No		No		No		No		Yes	

Note: All models are estimated with NielsenIQ data, 2011–2021, all states with effective minimum wages. Inflation is defined as the year-over-year change in the BLS CPI value. Income, education, household size, and age are from NielsenIQ Consumer Panel (HMS) data. Coefficient on ln(Minimum wage) is interpreted as the “pass-through elasticity” or the percentage change in average retail prices for a percentage change in the wage. Unit of observation is the retailer-county week. Data set consists of all retailers, in all counties, regardless of the number of competitors. Model 1 consists of controls, retailer, and week fixed effects; Model 2 adds retailer trends; Model 3 uses retailer-county fixed effects; Model 4 is Model 3 with county trends; and Model 5 uses retailer-county trends. Standard errors are clustered at the retailer level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

where s_t is the vector of retailer market shares in market t , x_t is a vector of retailer attributes, p_t are endogenous prices, and Ψ is the vector of demand parameters we wish to estimate. If $C_t(\Psi)$ is a market-specific constant, then ω_{jt} is a market-state vector composed of all the elements relevant to market competition, or $\omega_{jt} = \{s_{kt}, d_{j,kt}^x, d_{j,kt}^p\}$ such that $d_{j,kt}^x = x_{jt} - x_{kt}$ and $d_{j,kt}^p = p_{jt} - p_{kt}$ are the distances between firms j and k in attribute and price space, respectively. Distances cannot be used directly, however, as they are not likely to be independent of market-specific demand shocks, so we follow Miravete et al. (2022) and define distances instead using fitted prices from hedonic price regressions of the form

$$p_{jt} = \gamma_0 + \gamma_1 x_{jt} + \gamma_2 w_{jt} + v_{jt}, \quad (9)$$

where x_{jt} are exogenous store attributes and w_t are market-level cost shocks. In our application, we define the set of store attributes as the variety on offer each week (calculated as a simple sum of the number of stock-keeping units sold that week, as in Richards and Hamilton, 2015), the percentage of items on display, the percentage of items featured that week, and the number of stores owned by each retailer in the relevant market. We calculate cost shocks as price indices for retail wages, a producer price index of manufactured food prices, and indices of electricity, marketing, and transportation costs. All these latter variables are from the Bureau of Labor Statistics, while the retailer attributes are calculated from our NielsenIQ data. Once we estimate the model in (9), we then form fitted values and calculate distances using

$$z_{jt} = \sum_{r=1}^R \left(\frac{p_{rt}^{94} - p_{jt}^{94}}{p_{rt}^{94}} \right)^2 \quad (10)$$

for retail prices. Because these instruments reflect only exogenous variation in attributes and cost shocks, they are themselves exogenous and are not subject to the usual criticism of Hausman instruments (Hausman et al., 1994). Intuitively, expressing them as distances from all other prices in the market removes the part that is common to all retailers in the same market. We then use these instruments in a control-function framework in our random-parameters logit model described in the next section.

4 | EMPIRICAL MODEL

In this section, we describe the empirical model that we use to examine the effects of curvature, market power, and price inflation on minimum wage pass-through. Our approach is structural in the sense that we model how retail markets arrive at new equilibrium prices following a cost shock. If retailing costs change substantially due to an exogenous shock—such as from a change in minimum wages—retailers will each alter their prices in a way that reflects the structure of demand (including both the elasticity and curvature of demand), the nature of oligopoly rivalry among them, and how consumers search for what they believe to be the lowest price. Therefore, our empirical approach accounts for three factors: (1) the structure of demand through a discrete choice model of store choice, (2) oligopolistic price competition among retailers through a Bertrand-Nash model of retail pricing, and (3) consumer search behavior that modifies our Bertrand-Nash pricing model by assuming consumer search behavior changes as the rate of general price inflation rises. Because minimum wages only change retail prices through this equilibration process, we measure the rate of pass-through by simulating equilibrium responses to minimum wage-induced cost shocks and then decompose the importance of each factor through ex post regressions of the rate of minimum wage

pass-through on measures of curvature, market power, and general price inflation. We explain each element of this process in more detail in the empirical model description next.

4.1 | Demand model

We base our demand model on the unit-demand, discrete choice model of Miravete et al. (2022). We begin with a standard unit-demand, discrete choice model of market-level demand in which the object of demand is a grocery store visit (Berry & Haile, 2016). Grocery stores are highly differentiated in each local market, and we assume consumers choose one store from among many alternatives for their primary store visit during each purchase cycle. Although there is a large literature that considers the role of categories that may be particularly attractive at one store or another as a driver of store choice (Bell & Lattin, 1998; Briesch et al., 2009; Briesch et al., 2013), we abstract from this element of the consumer choice process as we focus only on store choice and the attributes of each store, and not the characteristics of each category in the store.¹⁸

We follow Miravete et al. (2022) and consider a general discrete choice model that is capable of producing both flexible substitution patterns and flexible curvature estimates, as both are important to the pass-through question, but we extend their model to recognize that search is also likely to affect elasticity and curvature. To that end, we write the utility function for consumer i in market t for store j as

$$u_{ijt} = \mathbf{X}_{jt}\beta + f_i(y_i, \pi_t, p_j) + \xi_j + \varepsilon_{ijt}, i \in I, j \in J, \quad (11)$$

where we assume ε is Type I extreme value distributed, inflation (π_t) varies by market, and the sub-utility function $f_i(y_i, \pi_t, p_j) = \alpha_i(y_i, \pi_t, \varphi_i)p_j = (\alpha_0 + \alpha_1 y_i + \alpha_2 \pi_t + \varphi_i)p_j$ captures the importance of income for demand curvature from Miravete et al. (2022) of inflation-driven search from Sara-Zaror (2024). We assume the outside option consists of spending on stores that are not modeled in our data, that are not in the NielsenIQ data, and assign a utility value of $u_{0ijt} = \varepsilon_{0ijt}$. We define the size of each market as the entire population of the county, multiplied by an average of 28 grocery items per week (the mean per capita in our data), multiplied by the average item price (BLP; Nevo, 2001). The importance of income is particularly true in our setting, as income is a primary determinant of whether a consumer visits a high-end supermarket or a lower-priced alternative, so it is likely to also be important to both the elasticity of demand, and its curvature. Inflation is also likely to be important, as the rate of general price inflation was relatively low in the early months of our data, but it rose substantially toward the end. Our specification also captures the specific type of mixing distribution that will both avoid generating positive demand elasticities, and a general pattern of curvature with the least empirical restriction. Namely, we allow the interaction parameter φ_i to follow a normal distribution ($\varphi_i \sim N(0, 1)$). With estimates from Equation (11), we recover elasticity and curvature estimates that vary over all observations in our data set using the approach described in Equations (6) and (7) above, and account for other features of our model (e.g., market power) using the approach in the next subsection.

4.2 | Pricing model

We extend the base model of Miravete et al. (2018, 2022) by allowing for the possibility that pass-through changes with both curvature and the general rate of inflation. Our consumer search

¹⁸Zhang et al. (2017) develop the concept of “polygamous store loyalty” that captures the notion that consumers may frequent multiple stores, depending on their needs and category attributes of stores in their choice set. We do not deny the validity of the polygamous store loyalty concept, but we maintain that consumers still have a preference for one store for the bulk of their grocery purchases.

hypothesis implies that retailers with market power pass through higher costs in inflationary environments because consumers are searching more actively (Sara-Zaror, 2024), and yet all pass-through outcomes are conditioned by the curvature of demand. Although curvature affects equilibrium pricing only implicitly through the matrix of own- and cross-price derivatives below, our search hypothesis maintains that inflation affects pricing both through its effect on demand as consumers search for low-priced products, and through pricing as firms respond to consumers' inability to search effectively. Both elements mean that pass-through varies over time and across retail markets.

We assume retailers maximize profit by choosing store-level prices under price (Bertrand-Nash) rivalry. We depart from the usual Bertrand-Nash pricing literature, however, by examining how retailers respond in equilibrium to a cost shock, in particular an increase in the minimum wage. In this model, the minimum wage rises, and retailers change prices in response to the cost shock in a way that restores the Bertrand-Nash price equilibrium, conditioned on how they expect consumers to respond to the change in prices, and how their rivals respond to consumer search activity. In this way, we capture the impact of imperfect competition, demand curvature, and inflation on minimum wage pass-through. We begin by assuming the profit function for retailer j is written as

$$\Pi_j = Ms_j(p_j - c_j) - F_j, \quad (12)$$

where Π_j is the profit for retailer j ; s_j is the market share of retailer j in its local market (county), from the expression preceding Equation (6) above; M is the total size of the local market (in volume); c_j is the unit retailing cost of selling groceries in retailer j ; and F_j reflects the retailer's fixed cost of selling items in all categories, suppressing the weekly time subscript for clarity. The unit cost of stocking items in Equation (12), which reflects both the costs of producing and retailing items in all categories, is a linear function of input prices, which results in the following expression for retailing costs:¹⁹

$$c_j(\mathbf{v}_j) = \sum_{l \in L} \eta_{wl} v_{jl} + \epsilon_j, \quad (13)$$

where $\mathbf{v}_j$ is a vector of L input prices, η_{wl} are estimates of the contribution of each input price to unit costs, and ϵ_j is an iid error term.

For our purposes, we assume retailing wages are separable between non-minimum wage employees and minimum wage employees. Assuming separability is necessary to obtain a pass-through estimate that is unique to minimum wage workers. We estimate retailing costs after substituting Equation (13) into the first-order conditions derived below. Conditional on the structure of demand, store j 's first-order condition for optimal prices becomes

$$\frac{\partial \Pi_j}{\partial p_j} = Ms_j + M \sum_{k \in J} (p_k - c_k) \frac{\partial s_k}{\partial p_j} = 0, \forall j, k \in J, \quad (14)$$

where $\partial s_k / \partial p_j$ is one element of a vector of store-sales derivatives with respect to price for all retailers in each market. Stacking the first-order conditions across retailers in each market, and

¹⁹We assume linear marginal cost is common in the empirical industrial organization literature (Kim & Cotterill, 2008; Villas-Boas, 2007). In this expression, fixed costs represent elements that do not vary with the volume of items sold, such as rent and management salaries. The marginal costs of retailing, however, includes minimum wage labor through restocking, checkouts, bagging, and other volume-related activities. We examine the sensitivity of our cost estimates to the assumption of linearity and find that the cost elasticity of both retailing wages and minimum wages varies little between the linear assumption, and either logarithmic or exponentiated cost indices.

suppressing the market notation for clarity, we write the Bertrand-Nash price equilibrium condition as

$$\mathbf{p} = \mathbf{c} - (\theta_j \mathbf{S}_p)^{-1} \mathbf{S} + \mathbf{X}\beta + \epsilon, \quad (15)$$

where bold notation indicates a vector (or matrix), $\mathbf{S}_p$ is the matrix of store-volume derivatives with element $\partial s_k / \partial p_j$, and $\mathbf{X}\beta$ are retailer and time-period fixed effects. We include the conduct parameter $2\theta_j$ consistent with the theoretical framework above to measure the extent of deviation from the maintained form of the pricing game, and to capture the effect of market power on pass-through. If our estimate is $2\theta_j = 1$, then retailers compete as Bertrand-Nash rivals, and any markup is due entirely to the extent of store differentiation reflected in the matrix of store-volume derivatives. If, however, we find $2\theta_j = 0$, then margins are zero and retail prices are consistent with perfect competition. Importantly, we allow our estimate of $2\theta_j$ to vary randomly over each observation in the data set, so $2\theta_j = \theta_0 + \theta_1 \tau$, where $\tau \sim N(0, 1)$, so our estimate of market power reflects unobserved variation in competitiveness by market and period.

To this point, our Bertrand-Nash equilibrium pricing specification is entirely conventional and consistent with others in the pass-through literature (Miravete et al., 2022, 2023). However, our consumer search hypothesis maintains that search manifests not only in the structure of demand, as explained above, but also in how retailers respond to the fact that all of their rivals understand that consumers may be impeded in their search activities due specifically to the fact that all prices are rising. We capture this hypothesis in the simplest way possible, namely, by assuming equilibrium prices respond to consumer search indirectly (through the $\mathbf{S}_p$ matrix above) and directly as an additional consideration in how they set prices. We test for this direct effect on retailer pricing by modifying Equation (15) above as

$$\mathbf{p} = \mathbf{c} - (\theta_j \mathbf{S}_p)^{-1} \mathbf{S} + \delta 2\pi + \mathbf{X}2\beta + \epsilon, \quad (16)$$

where 2π is our time-varying measure of inflation, and δ is the direct effect of search on retail prices. In the next subsection, we explain how we use the model in Equation (15) to simulate equilibrium price responses to cost shocks that emanate from changes in the minimum wage, and therefore estimate equilibrium pass-through rates.

4.3 | Simulating pass-through

Estimating equilibrium pass-through rates with our model in Equation (15) is not as simple as estimating reduced-form pass-through rates as in our Data section above. Equation (15) provides an expression for equilibrium prices, but prices are also arguments of the $\mathbf{S}_p$ matrix and $\mathbf{S}$ vectors, so they appear on both sides of the pricing model. Therefore, we find equilibrium pass-through rates using the following algorithm: First, we shock minimum wages by 100% and use the estimated marginal cost parameter on minimum wages from Equation (15) to find a “first pass” estimate of the change in retail prices.²⁰ Second, we use this estimated price change in the demand Expression (11) to calculate a new vector of market shares ($\mathbf{S}$) for each market and time period, and a new matrix of market share derivatives ($\mathbf{S}_p$). Third, the new share and share-derivative values allow us to calculate a “second pass” estimate of the equilibrium price, which we then feed back into the demand model. We iterate in this way until convergence and then define the price vector at convergence as the new

²⁰We use a 100% change in minimum wages to remain consistent with others in the structural pass-through literature (Miravete et al., 2023) but conduct our analysis for a range of assumed minimum wage changes and find the pass-through rate to be nearly identical regardless of the magnitude of the change.

equilibrium price vector. Fourth, we calculate the percentage change between the new and original equilibrium price vector and divide by our assumed 100% change in minimum wages to arrive at an estimated minimum wage pass-through rate, adjusted, of course, for the fact that wages only account for 17% of retail costs. Our approach is well accepted in the empirical industrial organization literature (Miravete et al., 2023) and is consistent with how industrial organization economists calculate price changes in response to mergers, new product introductions, or product removals (Conlon & Mortimer, 2021). Because the change in minimum wages is set at 100%, the calculated pass-through value is easily comparable to others in the literature (Leung, 2021; Renkin et al., 2022) who estimate reduced-form pass-through elasticities.

More importantly, this procedure generates a vector of pass-through rates that varies over all observations in the data set—over time, stores and retail markets. Because curvature (ρ) varies through the market shares estimated in demand Equation (11) as shown in Equation (7), market power (θ) varies across all retailers, markets, and time periods through our random parameter assumption in the pricing equation, and the general rate of inflation varies by month over the data set, we are able to decompose the simulated pass-through rate into parts that vary with curvature, market power, and the general rate of inflation. In this way, we test our core hypotheses regarding the relative importance of each different driver of retail pass-through with respect to changes in the minimum wage. We report the results of this exercise in the next section.

5 | RESULTS

In this section, we present and interpret our empirical results, beginning with estimates of the demand model and then moving to our findings from the structural pricing and simulation exercises.

5.1 | Demand estimates

We begin by presenting and explaining our demand estimates and then move to their implications for pricing and pass-through in the next subsection. Estimates from different versions of our demand model are in Table 4. In this table, we show estimates from successive versions of the model in which we control for different levels of fixed effects (Models 1 and 2), adding a control for endogeneity in Model 3, and allowing for a fully flexible marginal utility of income in Model 4. Because the fully saturated model (Model 4) provides the best fit to the data ($\chi^2 = 753.02$ relative to Model 3), we interpret our findings using this model.

There are several estimates of note in Table 4, but we will focus on those that are of interest to our objective of estimating pass-through, and elements of the economic environment that affect pass-through. First, note that the income effect in the random-price parameter is significant, and including income as a measure of price response heterogeneity explains a considerable amount of demand variation between stores and over time. Therefore, our model is likely flexible enough to capture the sort of elasticity and curvature variation noted by others in the literature (Birchall & Verboven, 2022; Miravete et al., 2022, 2023). Exploiting this flexibility, we estimate price elasticity and curvature values for every observation in the data set using Equations (6) and (7) above, respectively. Across the entire data set, the estimates in Table 4 imply an average own-price elasticity of -2.422 (standard deviation = 0.691), which is relatively elastic compared to other store choice models but likely reflects the fact that our model accounts more completely for factors that likely cause retail prices to be endogenous.²¹ Second, we find an average curvature value of 1.0239

²¹It is well understood that the direction of bias in estimating demand elasticities is typically positive, so removing sources of endogeneity leads to larger, or more negative, price elasticities of demand.

T A B L E 4 Mixed logit store demand model estimates.

	Model 1		Model 2		Model 3		Model 4		Model 5	
	Est.	Std. err.	Est.	Std. err.	Est.	Std. err.	Est.	Std. err.	Est.	Std. err.
Constant	−7.7203***	0.0598	−4.1835***	0.0469	−7.2662***	0.0634	−7.0568***	0.0592	−7.0740***	0.0585
Feature	2.3220***	0.1569	0.3127***	0.0232	0.3322***	0.0219	0.2903***	0.0207	0.2946***	0.0208
Display	0.6345***	0.1190	−0.2690***	0.0142	−0.2386***	0.0168	−0.2280***	0.0154	−0.2289***	0.0154
Variety	0.1318***	0.0018	0.1573***	0.0018	0.1383***	0.0016	0.1401***	0.0015	0.1402***	0.0015
Num. stores	0.5323***	0.0056	0.4008***	0.0090	0.7516***	0.0053	0.6740***	0.0058	0.6738***	0.0058
CF × Price	NA	NA	NA	NA	0.0575***	0.0014	0.0564***	0.0014	0.0548***	0.0014
Price	0.0002	0.0141	−0.1210***	0.0058	−0.4400***	0.0095	−0.1611***	0.0109	−0.1502***	0.0111
Price(Inc)	NA	NA	NA	NA	NA	NA	−0.0041***	0.0010	−0.0041***	0.0010
Price(Inf)	NA	NA	NA	NA	NA	NA	NA	NA	−0.1939***	0.0302
Weeks?	No		Yes		Yes		Yes		Yes	
Years?	No		Yes		Yes		Yes		Yes	
Retailers?	No		Yes		Yes		Yes		Yes	
N	17,343		17,343		17,343		17,343		17,343	
LLF	−32,648.13		4509.52		4847.03		5188.69		5197.27	
AIC	3.7657		−0.5139		−0.5511		−0.5902		−0.5911	

Note: All models were estimated with Kilts/RMS store-level scanner data for a seven-state sample of minimum wage and non-minimum wage states. Note that retailers and geographic markets are not separately identified, so we report retailer-level fixed effects. Model 1 is a simple logit model with only attributes and store-price indices. Model 2 adds fixed retailer, week, and year fixed effects; Model 3 adds a control function; Model 4 adds random parameters and an interaction of the marginal utility of income with a household-income draw; and Model 5 adds search behavior, proxied by interacting the marginal utility of income with the general level of price inflation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

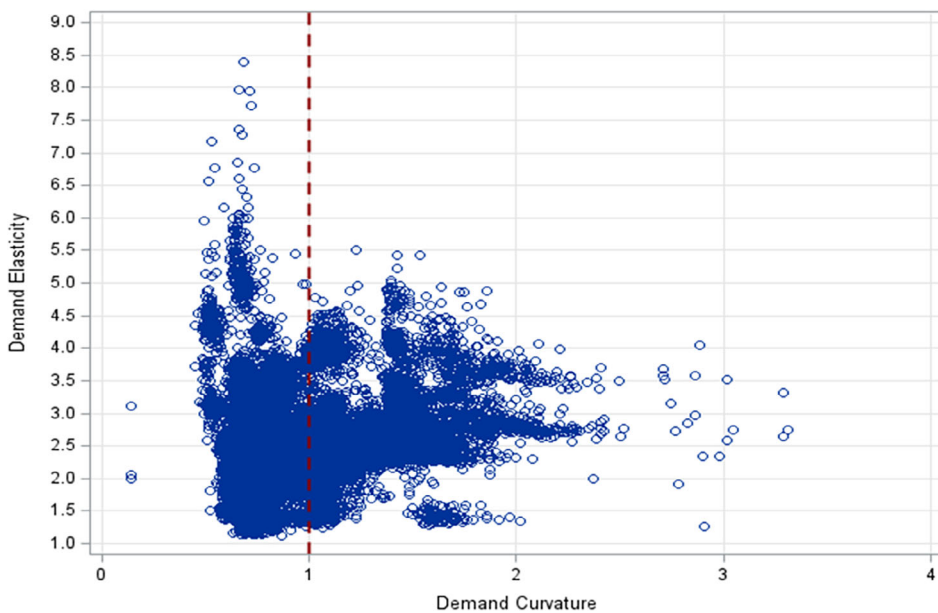


FIGURE 5 Demand manifold, mixed logit model of store choice, various U.S. retail markets, 2011–2021. Estimated with 2011–2021 NielsenIQ/Kilts store choice data over seven sample U.S. markets, on a federal information processing standard (county) and weekly level. Store choice elasticities and curvature were estimated using expressions in Miravete et al. (2022). Elasticities and curvature values are estimated for each point in the data and are not averages.

(standard deviation = 0.3514), which is slightly log-convex. Figure 5 shows the demand manifold implied by these curvature and elasticity estimates, and it implies a significant number of elasticity and curvature pairs well into the sub-convex region of the manifold. Consequently, we expect to find an average pass-through rate that is greater than 1.0, or minimum wages that are overshifted into retail prices.

Among the other parameters reported in Table 4, we note that variety and breadth of retail distribution have strong and statistically significant effects on retailer-level market share in each geographic market. Neither of these results are surprising (Bell & Lattin, 1998; Briesch et al., 2009), but they point to the fact that our specification controls for factors that are otherwise likely to explain variation in market share and would otherwise confound our elasticity and curvature estimates if not accounted for. The negative estimate for the display variable is somewhat surprising, but not in a model in which we already control for the demand-increasing effect of featuring a product and price variation.

5.2 | Retail pricing model results

Here, we discuss our pricing model estimates and their implications for how curvature and inflation affect the rate at which changes in minimum wages are passed through to consumers. Recall that our pricing model assumes retailers compete as Bertrand-Nash oligopolists in their local market, conditioned on the structure of demand shown in Table 4. In this model, curvature is only implicit, so we use the pricing model to estimate the marginal cost of retailing labor and then use counterfactual simulation to solve for equilibrium prices that result following a shock to minimum wages. We interpret the resulting change in prices as the extent to which higher minimum wages are passed through in each market.

We show our pricing model estimates in Table 5. We estimate three versions of the pricing model in Equation (15), accounting for three different fixed-effect structures in addition to two

TABLE 5 Bertrand-Nash pricing model estimates.

	Model 1		Model 2		Model 3	
	Est.	Std. err.	Est.	Std. err.	Est.	Std. err.
Constant	−4.8170***	0.2215	2.7650***	0.1082	2.6716***	0.1091
Retail wage	0.0108***	0.0004	0.0062***	0.0001	0.0070***	0.0002
Minimum wage	0.0887***	0.0077	0.0281***	0.0027	0.0191***	0.0027
Inflation	−0.3253	0.5023	1.6456***	0.1378	1.1332***	0.1390
Conduct	−2.2972***	0.0961	0.6782***	0.0404	0.6658***	0.0390
Month fixed effects	No		No		Yes	
Retailer fixed effects	No		Yes		Yes	
LLF	115.4132		116.4251		726.6603	
AIC/N	−0.0081		−0.0082		−0.0771	

Note: All pricing models were estimated using markups implied by estimates in Table 4. Retail wages are from the U.S. Bureau of Labor Statistics, and minimum wages are from sources reported in Table 3. Model 1 is the base model with only retail wages, minimum wages, and markup term; Model 2 adds retailer fixed effects; and Model 3 adds monthly fixed effects. Conduct parameter was allowed to vary randomly over data in all three models. All models estimated with standard errors clustered at the market level. *** $p < 0.010$, ** $p < 0.05$, * $p < 0.10$.

measures of labor-based retailing costs and the search effect, as proxied by the general rate of inflation. Our preferred model is a two-way fixed-effects model in which we account for fixed effects for the months of the data set and each individual retailer.²² Because the marginal cost function is linear, each of the cost estimates is interpreted as contributing to the unit cost of producing groceries. Both the retail wage index and minimum wages are statistically significant and positive, which is to be expected.²³ These parameters, however, cannot be used to interpret pass-through directly, as equilibrium markups are likely to change following a shock to any cost item, which will change both the left side and right side of the model in Equation (15). Consequently, estimating pass-through involves re-solving for equilibrium prices conditional on the nature of oligopolistic rivalry in the market. In this regard, our conduct parameter estimate ($\theta = 0.666, p < 0.01$) implies that retailers in our sample, on average, are more competitive than the maintained Bertrand-Nash assumption ($\theta = 1$) but are not perfectly competitive ($\theta = 0$).²⁴ When firms are imperfectly competitive, the interaction between curvature and market conduct means that pass-through is an empirical question (Miller et al., 2017), and overshifting costs is a very real possibility.²⁵

We summarize our findings regarding the pass-through of a shock to minimum wages in Table 6. We follow the experiment in Miravete et al. (2022) by allowing minimum wages to change by 100%, or doubling. Although a change in minimum wages of this magnitude is unrealistic, the size of the change does not affect the relative magnitude of the change in retail prices, and starting with a 100% change allows for a clean interpretation of the result.²⁶ Based on our assumed change in minimum wages, the estimates in Table 6 show that we expect retail prices to rise by roughly 5.2%

²²Because retailer banners are generally unique to each geographic market, it was not possible to include market-level fixed effects as well.

²³Controlling for differences in the magnitude of each cost variable, the cost elasticity with respect to retailing wages is 1.047 and with respect to the minimum wage is 0.069. In this regard, retailing wages have a larger impact on retail prices, as expected.

²⁴We acknowledge that estimating conduct parameters is controversial (Corts, 1999) but follow Weyl and Fabinger (2013) in interpreting conduct as a parametric estimate of the net effect of accounting for all diversion ratios across all retailers in each market. Weyl and Fabinger (2013) show that allowing for conduct is essential in modeling pass-through in imperfectly competitive industries.

²⁵We examine the robustness of our assumptions in how we model cost by considering log and exponentiated versions of the cost equation, and our minimum wage and conduct estimates are very similar to those shown in Table 5. We also account for the possibility that minimum wages have a “ripple effect” throughout the wage distribution, as in Autor et al. (2016) and Neumark et al. (2004), by breaking the retail wage into wages at the 10th and 50th (median) percentiles from the Current Population Survey and again find very little difference in our cost and conduct parameter estimates. These robustness checks are available from the authors.

²⁶We also estimate the model assuming a 50% change in the minimum wage and a 10% change, and our pass-through estimate is nearly identical. This finding also suggests that the potentially important second-order effect of a change in the minimum wage of this size on household income, and hence demand, is too small to be of any practical importance. These results are available from the authors.

TABLE 6 Summary of pricing and pass-through estimates.

Variable	Base value		New equilibrium	
	Est.	Std. err.	Est.	Std. err.
Minimum wage	9.2339	1.8642	18.4678	3.7284
Curvature	1.0239	0.3514		
Elasticity	−2.4156	0.6940		
Retail price	3.5575	1.0113	3.7300	1.0160
Markup	1.4783	0.1238	1.4728	0.1250
Market share	0.0062	0.0101	0.0054	0.0089
Pass-through (%)			0.0515	0.0153

Note: All estimates are based on flexible demand-model estimates in Table 4. All estimates are at sample-average values. Minimum wage simulation shocks wage by 100% as in Miravete et al. (2023), so pass-through percentage value is also interpreted as an elasticity. Robustness checks for a range of minimum wage shocks show that this value holds regardless of the simulated change in minimum wages.

in equilibrium, with a slight reduction in markups (−0.40%). Although media reports during the inflationary period of 2021–2022 interpreted this outcome as “greedflation,” we show that it is indeed a natural outcome of oligopoly retailers passing through costs in market equilibrium. The fact that markups fall slightly following a change in costs is more a function of the relatively competitive nature of our sample markets than it is to curvature because log-convex demand means that retailers have an incentive to raise prices for high-willingness-to-pay consumers, at the risk of losing low-willingness-to-pay consumers, as the increase in profit from serving the former outweighs the costs of losing the latter. The fact that the net outcome—a reduction in margins—shows the opposite means that curvature is not the dominant influence on pass-through.²⁷

In fact, if we allow for the fact that 25% of retail grocery workers earn the minimum wage (calculated from the U.S. Current Population Survey), and that labor accounts for some 14% of retail operating costs (National Grocers Association, 2021; U.S. Census Bureau, 2023), our estimates imply a pass-through rate of some 147.3%, or well above the level of complete cost pass-through.²⁸ Controlling for the structure of demand, market power in each retail market, and other labor costs, therefore, changes in the minimum wage are overshifted into retail prices.

Our estimate is higher than in Renkin et al. (2022, 3.6%) and slightly lower than the relatively high estimate of Leung (2021, 7.0%), but neither of these studies account for the equilibrium change in prices and the effect of market power on cost pass-through. Reduced-form estimates of pass-through cannot control for the impact of market power on retail pricing because conduct in imperfectly competitive markets is endogenous in a way that cannot be addressed by simply adding fixed effects or even conventional instrumental-variables strategies (Miller et al., 2022). Further, neither of these studies consider pass-through during inflationary periods, which may be substantially different because consumers expect higher prices when all prices are rising. In a high-inflation environment, the possibility of using a general increase in prices to pass through cost increases more effectively is a real possibility, as it is a consequence of consumer search behavior (Byrne & de Roos, 2017; Lewis, 2011) or the ability to profit from confusion.²⁹ Disentangling the effect of inflation from the structure of demand directly, however, is not possible, so we conduct a post-simulation regression analysis to disentangle the relative contribution of each.

²⁷The results in Table 6 also show a slight decrease in the mean market share among our inside, or modeled, stores. This is to be expected, as an increase in retail prices among the supermarkets in the NielsenIQ/Kilts data is likely to lead to substitution into stores in the outside option, such as dollar stores and deep discounters.

²⁸Pass-through is roughly 147.3%, as complete pass-through would mean that a 100% change in wages leads to a change in prices of $0.25 \times 0.14 = 3.5\%$. Our estimate of 5.2% is 147.3% of this value.

²⁹This notion paraphrases a statement by Tony Curtis in the movie *Petticoat Junction* but refers to a more general social-psychological concept.

As we discuss above, the effect of curvature on pass-through is implicit in the Bertrand-Nash pricing model developed in the previous section, inflation and search enter through both the demand and pricing equilibrium models, and market power enters only through the pricing model. Given that each element affects pass-through in a different way, we examine the relationship between curvature, market power, and inflation with pass-through using a set of secondary two-way fixed-effects regression models in which we use variation in each to explain variation in pass-through over time and across our sample retailers.³⁰ We show our estimates from these two-way fixed-effects regressions in Table 7, first estimating each effect on their own for curvature (Model 1), inflation (Model 2), and market power (Model 3), and then combining all three in Model 4. Because Model 4 provides the best fit to the data and controls for correlation among each factor, we interpret our findings from this model. As expected, each variable is positively related to pass-through rates over time and across retailers, although they vary in their importance and strength of relationship to pass-through.

Based on the estimates in Table 7, pass-through depends to a much greater extent on market power than either curvature or search behavior. In fact, the estimates in Table 7 suggest that a change in market power from the sample mean of $\theta = 0.666$ to an even modestly less competitive level that is 10% closer to Bertrand-Nash competition (or $\theta = 1.0$) can be expected to produce a pass-through rate that is fully 74% higher, or from the sample mean of 5.1% to nearly 8.5%. In practical terms, even a small increase in market power, therefore, has the potential to change a situation in which costs are slightly overshifted into retail prices into one in which costs are clearly overshifted. In terms of the policy debate regarding the nature and causes of “greedflation,” our finding supports the notion that firms with more market power pass along higher costs—wages, in this case—simply because they can.

Curvature and consumer search behavior, as proxied by the rate of inflation, remain important to the equilibrium pass-through rates. With respect to curvature, the estimates in Table 7 imply that a 1-point increase in curvature (not measured in percentages), or roughly from the mean to the upper 95th percentile in our data based on the demand manifold in Figure 5, implies a 0.8 percentage point increase in pass-through (instance.g., from 5.0% to 5.8%). Figure 6 shows this relationship in graphical form and highlights the fact that something other than curvature seems to explain differences in the pass-through rate, particularly at low levels of curvature. Based on the theoretical model of pass-through from Weyl and Fabinger (2013) in Equation (2) above, the reason is clear: curvature only mediates the effect of market power on pass-through, so when retailers play something near Bertrand-Nash ($\theta = 1$) and curvature itself floats around 1.0, the entire second term in the theoretical pass-through equation nearly disappears. In fact, when we restrict our sample to only high-curvature observations, the estimated effect of pass-through rises from 0.4 to 1.5, so curvature becomes much more important for pass-through in some markets.³¹

Inflation is an order of magnitude less important than the other two factors, yet still positively related to pass-through. The estimates in Table 7 show that a 1 percentage point increase in inflation (e.g., from 1.0% rate of increase in prices to 2.0%) results in a roughly 0.2 percentage point increase in pass-through (e.g., from 5.0% to 5.2%). Because inflation reached a peak of some 7.0% in our sample (2021) and more than 9.0% during 2022 (see Figure 1), our parameter estimates suggest that the inflation effect is potentially large enough to change an undershifting environment, in which retailers absorb cost increases and suppress food price inflation, to an overshifted environment in which cost increases are magnified and the appearance of greedflation sets in. Still, the inflation effect is much smaller, proportionately, than the effect of market power.

Sensitivity, or the marginal effect of each variable, is one measure of the relative importance of each variable, but we can also decompose the relative importance of each effect in a statistical sense by calculating the partial coefficients of determination associated with each pass-through

³⁰Note that we use the national rate of inflation in these regressions and measure pass-through at the individual store level, so any endogeneity through individual store prices is likely to be very small.

³¹The estimates when we restrict our sample to only $\rho > 1$ observations are available from the authors.

T A B L E 7 Pass-through relationship with curvature and inflation.

	Model 1		Model 2		Model 3		Model 4		Partial R ²
	Est.	Std. err.	Est.	Std. err.	Est.	Std. err.	Est.	Std. err.	
Constant	0.0314***	0.0009	0.0471***	0.0009	−0.3375***	0.0064	−0.4822***	0.0558	
Curvature	0.0138***	0.0002	NA	NA	NA	NA	0.0085***	0.0020	20.43%
Inflation	NA	NA	0.0866***	0.0037	NA	NA	0.1785***	0.0298	2.88%
Conduct	NA	NA	NA	NA	0.5839***	0.0096	0.7207***	0.0831	21.32%
Month fixed effects	Yes		Yes		Yes		Yes		
Retailer fixed effects	Yes		Yes		Yes		Yes		
R ²	0.8503		0.8232		0.8499		0.8849		
F	1791.31		1468.63		1786.57		2340.02		

Note: All models were estimated using pass-through estimates from the Bertrand-Nash pricing model in Table 5. Curvature was estimated from the demand model using expressions from Miravete et al. (2023). All models were estimated with standard errors clustered at the market level. Model 1 examines the curvature effect in isolation, Model 2 the inflation effect, and Model 3 the conduct or market power effect. Model 4 combines all three. Partial R² shows the percentage variation in pass-through explained by each variable. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

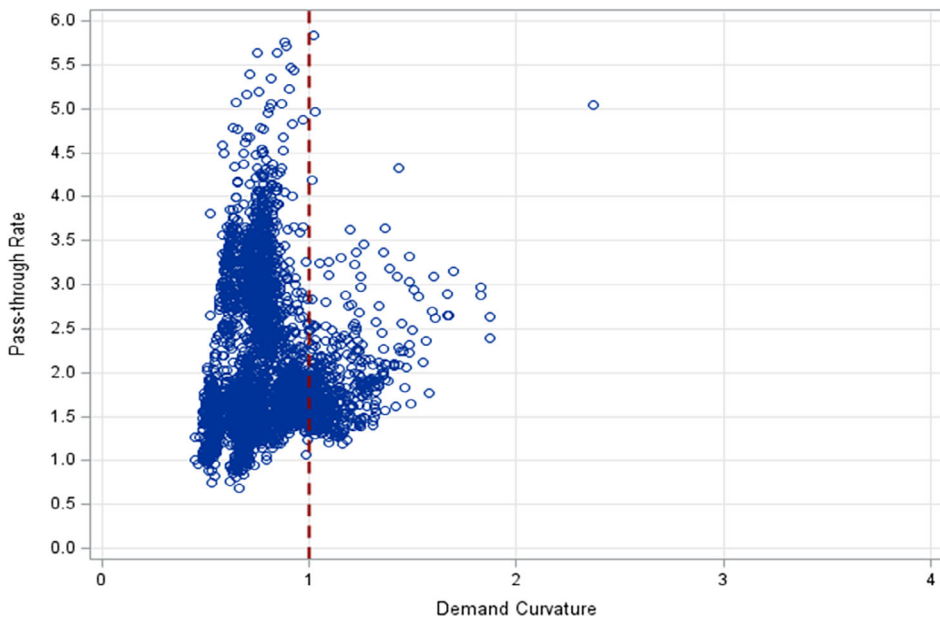


FIGURE 6 Demand curvature and pass-through rates, various U.S. retail markets, 2011–2021. Estimated with 2011–2021 NielsenIQ/Kilts store choice data over seven sample U.S. markets, on a federal information processing standard (county) and weekly level. Pass-through values were simulated on the assumption stores compete as Bertrand–Nash oligopolists in each local market, and prices reequilibrate after 100% shock to minimum wages. Pass-through values assume wages represent 14% of retail operating costs (U.S. Census Bureau, 2023). Curvature values were estimated from demand model estimates shown in Table 5.

determinant. In Table 7, these estimates answer the following question: How much of the relative variation in pass-through does each variable explain, controlling for the effect of all the others? Our estimates show that curvature and conduct, or market power, explain roughly similar amounts of variation in pass-through (20.4% and 21.3%, respectively), whereas inflation explains much less, at roughly 2.9%. Together, this decomposition shows that these structural factors explain almost half of the variation in retail price pass-through, whereas the other half is due to retailer, market, and time fixed effects, or idiosyncracies that cannot otherwise be explained through our model.

Our findings provide an important link in the policy discussion surrounding the effect of minimum wages on retail food prices, and on the causes of inflation more generally. First, we show that understanding the structure of demand is critical to understanding how changes in cost are likely to affect retail prices. Whereas traditional models of retail pricing, based on BLP-type demand models, cannot account for the effect of curvature on demand, by using a modeling approach that permits flexibility in both elasticities and curvature we are able to generate more accurate estimates of pass-through that address the effect of consumer behavior on retail pricing. Second, we provide empirical evidence that greedflation may indeed be present in retail pricing, through the exercise of market power, however, and not necessarily through retail concentration as described in the media.³² Although we cannot test our consumer search hypothesis directly, as others who use experimental data can (Allender et al., 2021), we find evidence that pass-through rises in periods of general price inflation. Indeed, our search hypothesis implies that retailers are able to use consumers' expectations of rising prices as a self-fulfilling outcome—if consumers expect higher prices, and retailers profit from their expectations, then prices will indeed rise. Third, we show that there is no bright line

³²High levels of retail concentration are not the same as the exercise of market power, particularly when retailers compete in prices (Weyl & Fabinger, 2013).

between market structure, or even market conduct, and pass-through as the transmission of costs to retail prices results from a complex interplay of the structure of demand, the general macroeconomic environment, and the extent of imperfect competition.

6 | CONCLUSION

In this article, we provide an empirical analysis of minimum wage pass-through among a sample of food retailing data using a structural approach in which we account for the exercise of market power, consumer search behavior, and demand curvature. We argue theoretically that pass-through is likely to differ in high-inflation settings, so the rise in inflation during 2021–2022 provides an ideal setting to examine how inflation, and consequent changes in consumer search, affects pass-through. Although there has been a considerable amount of recent research on this topic using data from previous periods of low price inflation (Leung, 2021; Renkin et al., 2022), we are the first to consider how pass-through differs across different regimes of fundamentally different experiences with general price inflation. Our hypothesis is that pass-through is higher when all prices are rising, and we base our expectations on a theoretical model of consumer search and consumer expectations.

We find that accounting for demand curvature and consumer search is important when estimating cost pass-through rates. However, our empirical findings suggest that accounting for market power is more important. In fact, our empirical estimates point to a market power effect that is more important than demand curvature alone, and an order of magnitude more important than search behavior. Although there is strong theoretical support for the idea that retailers should be able to pass through higher costs at a greater rate when all prices are rising—when consumers are unsure as to what the true price of an item is supposed to be—sellers can exploit this informational uncertainty to their advantage. This mechanism appears to be less important than market power empirically.

Our findings are important because much of the policy discussion surrounding the 2021–2022 period of food price inflation centered on retail concentration and the ability of monopoly retailers to manipulate retail prices. Although concentration and market power are not the same thing (Miller et al., 2022), we find that there is some empirical basis to the argument that the exercise of market power is associated with higher pass-through rates. To the extent that concentration represents the potential exercise of market power, we find that there may be some merit to the policy argument that monopoly power may contribute to food price inflation.

Future research in this area would benefit from more data during high-inflation periods. Because our retail data only capture the beginning of the inflationary period in 2021, subsequent empirical analyses may find something different when inflation appeared to peak in 2022 and declined thereafter. Further, we are confident that our demand model captures a wide range of possible curvature scenarios, but other research in this area (Birchall & Verboven, 2022; Miravete et al., 2022, 2023) points to a much larger set of tools that are able to account for demand curvature in applications like this. Additionally, we consider minimum wage pass-through as an important example of why pass-through matters in an economically critical industry, but there are other settings in which pass-through is important and worthy of investigation over the inflationary period of 2021–2022. Horizontal mergers, new-product introductions, and changes in distribution technologies are all examples of cases in which pass-through during inflationary periods is likely to differ from non-inflationary periods.

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of Business. The conclusions drawn from the NielsenIQ data are those of the researcher(s) and do not reflect the views of NielsenIQ. NielsenIQ is not responsible for, had no role in, and was not involved in analyzing and preparing the results reported herein.

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